

Piezoelectric Energy Harvester: Modeling Assumptions, ROM Derivation, and Slow-Flow Analysis

Working note for collaborators

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Purpose

This note organizes the reasoning path behind the piezoelectric energy harvesting model used in the weekly meeting slides and MATLAB simulations. It is not intended to be a polished paper derivation. The goal is to make the modeling choices, reduction steps, and analysis pipeline clear enough that a collaborator can check the logic and connect the equations to the simulation files.

Main sources. The derivation below is organized from the archived Word notes `Old_sources/piezoelectricbeamharvester.docx`, `Old_sources/ROM-sendtoMohammod.docx`, `Old_sources/ROM-multiple-sendtoMohammod.docx`, `Old_sources/Dampinganalysis.docx`, and the weekly meeting slides in `docs/weekly_meeting/`. The corresponding cleaned MATLAB archive is in `simulations/`.

1 Physical Picture and Assumptions

1.1 Beam and Electrical Paths

The physical system is a cantilever piezoelectric beam under prescribed base motion. Two piezoelectric electrical paths are used:

1. **Path 1: harvester path.** This path is connected to a shunted electrical circuit. It is the path from which harvested electrical energy is measured. The harvester circuit contains capacitance, resistance, linear inductance, and an effective cubic nonlinear inductive term.
2. **Path 2: modulator path.** This path is not solved as an autonomous circuit in the reduced model. Its voltage is prescribed and used as a controllable input. In the main simulations it is taken near the base excitation frequency so that it slowly modulates the effective forcing.

The first path is therefore a dynamical electrical degree of freedom. The second path is an externally imposed actuation/modulation signal.

1.2 Mechanical Assumptions

- The beam is modeled as an Euler-Bernoulli beam.
- The total transverse displacement is decomposed into base motion plus relative beam deflection:

$$w(x, t) = w_b(t) + w_{\text{rel}}(x, t). \quad (1)$$

- The base motion is prescribed. The archived notes often write $w_b(t) = g(t)$, and for harmonic studies use

$$w_b(t) = w_0 e^{j\omega_b t} + \text{c.c.} \quad (2)$$

- In the initial ROM derivation, structural damping of the beam is neglected. A later extension includes a modal damping ratio ξ .
- The first ROM uses one mechanical mode:

$$w_{\text{rel}}(x, t) = \sum_r \eta_r(t) \phi_r(x) \approx \eta_1(t) \phi_1(x). \quad (3)$$

- The mode shape is mass-normalized:

$$\int_0^l m \phi_1^2(x) dx = 1. \quad (4)$$

Important correction. The earliest ROM used the analytic first bending mode of a plain cantilever. The January 17 meeting identified that this approximation shifted the relevant electromechanical frequency enough to cause disagreement with the full beam simulation. The corrected ROM uses a mode computed from the FEM eigenanalysis with the piezoelectric/electrical coupling included. In the cleaned MATLAB archive this is the role of:

- `simulations/fem/eigenanalysis_corrected_single_mode.m`
- `simulations/fem/FEM_modes_norm.mat`
- `simulations/rom/corrected_fem_mode_rom.m`

2 From Beam-Circuit System to a Two-DOF ROM

2.1 Beam Moment and Governing Equation

The bending moment is modeled as the elastic moment plus two localized piezoelectric moment contributions:

$$M = -YI \frac{\partial^2 w_{\text{rel}}}{\partial x^2} + \vartheta v_1(t) \left[H(x) - H\left(x - \frac{l}{2}\right) \right] + \vartheta v_2(t) \left[H\left(x - \frac{l}{2}\right) - H(x - l) \right]. \quad (5)$$

Here v_1 is the harvester-path voltage and v_2 is the modulator-path voltage. The bending stiffness and piezoelectric coupling coefficient used in the archived notes are

$$YI = \frac{2b}{3} \left[Y_s \frac{h_s^3}{8} + c_{11}^E \left(\left(h_p + \frac{h_s}{2} \right)^3 - \frac{h_s^3}{8} \right) \right], \quad (6)$$

$$\vartheta_{\text{series}} = \frac{e_{31}b}{2} (h_p + h_s). \quad (7)$$

Neglecting beam damping in the main derivation gives

$$-\frac{\partial^2 M}{\partial x^2} + m \frac{\partial^2 w_{\text{rel}}}{\partial t^2} = -m \ddot{w}_b(t). \quad (8)$$

2.2 Single-Mode Projection

Projecting onto the first mass-normalized mode gives

$$\ddot{\eta}_1 + \omega_1^2 \eta_1 - \chi_{11} v_1 - \chi_{12} v_2 = f_1(t), \quad (9)$$

where

$$\omega_1^2 = \int_0^l YI \phi_1^{(4)}(x) \phi_1(x) dx, \quad (10)$$

$$\chi_{11} = \vartheta \int_0^{l/2} \phi_1''(x) dx = \vartheta \phi_1' \left(\frac{l}{2} \right), \quad (11)$$

$$\chi_{12} = \vartheta \int_{l/2}^l \phi_1''(x) dx = \vartheta \left[\phi_1'(l) - \phi_1' \left(\frac{l}{2} \right) \right], \quad (12)$$

$$f_1(t) = -m\ddot{w}_b(t) \int_0^l \phi_1(x) dx. \quad (13)$$

The analytic plain-cantilever mode used in the initial notes is

$$\phi_r(x) = \frac{1}{\sqrt{ml}} \left[\cos \frac{\lambda_r x}{l} - \cosh \frac{\lambda_r x}{l} + \frac{\sin \lambda_r - \sinh \lambda_r}{\cos \lambda_r + \cosh \lambda_r} \left(\sin \frac{\lambda_r x}{l} - \sinh \frac{\lambda_r x}{l} \right) \right], \quad (14)$$

with

$$1 + \cos \lambda_r \cosh \lambda_r = 0. \quad (15)$$

In the corrected model, ϕ_1 , ω_1 , χ_{11} , χ_{12} , and $\int_0^l \phi_1 dx$ should be taken from the FEM-informed mode data rather than from the plain beam formula.

2.3 Harvester Circuit

Let λ_1 be the flux linkage in the harvester path so that $v_1 = \dot{\lambda}_1$. The harvester circuit equation is

$$C_p \ddot{\lambda}_1 + \frac{1}{R_1} \dot{\lambda}_1 + \frac{1}{L} \lambda_1 + \frac{1}{L_c} \lambda_1^3 + \chi_{11} \dot{\eta}_1 = 0. \quad (16)$$

Using $v_1 = \dot{\lambda}_1$, the mechanical equation becomes

$$\ddot{\eta}_1 + \omega_1^2 \eta_1 - \chi_{11} \dot{\lambda}_1 = \chi_{12} v_2(t) + f_1(t). \quad (17)$$

Equations (16) and (17) form the two-degree-of-freedom reduced system. The two solved DOFs are the structural modal coordinate η_1 and the harvester flux linkage λ_1 . The modulator signal $v_2(t)$ is prescribed.

3 Normalization

The archived notes use

$$\lambda_0 = \sqrt{L_c C_p \omega_1^2}, \quad \eta_0 = \lambda_0 \sqrt{C_p}, \quad t_0 = \frac{1}{\omega_1}, \quad T = \frac{t}{t_0}. \quad (18)$$

Define dimensionless variables

$$\Lambda_1 = \frac{\lambda_1}{\lambda_0}, \quad H_1 = \frac{\eta_1}{\eta_0}. \quad (19)$$

The dimensionless parameters are

$$\zeta = \frac{1}{C_p R_1 \omega_1}, \quad \Omega_t = \frac{\omega_t}{\omega_1}, \quad \omega_t^2 = \frac{1}{C_p L}, \quad \Theta = \frac{\chi_{11}}{\omega_1 \sqrt{C_p}}. \quad (20)$$

For harmonic base excitation and prescribed modulation voltage,

$$w_b(t) = w_0 e^{j\omega_b t} + \text{c.c.}, \quad v_2(t) = v_0 e^{j\omega_v t} + \text{c.c.}, \quad (21)$$

the normalized forcing amplitudes are

$$F_b = \frac{m\omega_b^2 w_0}{\omega_1^2 \eta_0} \int_0^l \phi_1(x) dx, \quad (22)$$

$$F_v = \frac{\Theta \chi_{12} v_0}{\chi_{11} \omega_1 \lambda_0}. \quad (23)$$

The main normalized two-DOF ROM is therefore

$$H_1'' + H_1 - \Theta \Lambda_1' = F_b e^{j\Omega_b T} + F_v e^{j\Omega_v T} + \text{c.c.}, \quad (24)$$

$$\Lambda_1'' + \zeta \Lambda_1' + \Omega_t^2 \Lambda_1 + \Lambda_1^3 + \Theta H_1' = 0, \quad (25)$$

where primes denote derivatives with respect to T and $\Omega_b = \omega_b/\omega_1$, $\Omega_v = \omega_v/\omega_1$.

Optional structural damping. For damping analysis, the structural equation is extended to

$$H_1'' + 2\xi H_1' + H_1 - \Theta \Lambda_1' = F e^{j\Omega_f T} + \text{c.c.}. \quad (26)$$

4 Slow-Flow Analysis for the Two-DOF System

4.1 Slowly Varying Effective Forcing

When the base and modulator frequencies are close, the forcing can be viewed as a carrier at $\Omega_f \approx \Omega_b$ with a slowly varying envelope:

$$F_b e^{j\Omega_b T} + F_v e^{j\Omega_v T} = \left(F_b + F_v e^{j(\Omega_v - \Omega_b)T} \right) e^{j\Omega_b T}. \quad (27)$$

For the frozen-envelope analysis, write this as

$$F e^{j\Omega_f T} + \text{c.c.}, \quad (28)$$

where F is treated as slowly varying relative to the carrier oscillation.

4.2 Complex Amplitude Variables

Define the complex amplitude variables

$$\psi_H = H_1' + j\Omega_f H_1, \quad \psi_\Lambda = \Lambda_1' + j\Omega_f \Lambda_1. \quad (29)$$

Then

$$H_1' = \frac{\psi_H + \overline{\psi_H}}{2}, \quad H_1 = \frac{\psi_H - \overline{\psi_H}}{2j\Omega_f}, \quad (30)$$

and similarly for Λ_1 . Assume

$$\psi_H = \varphi_H e^{j\Omega_f T}, \quad \psi_\Lambda = \varphi_\Lambda e^{j\Omega_f T}, \quad (31)$$

where φ_H and φ_Λ vary slowly.

Keeping only the resonant $e^{j\Omega_f T}$ terms gives the slow-flow system

$$\dot{\varphi}_H + \frac{1 - \Omega_f^2}{2j\Omega_f} \varphi_H - \frac{\Theta}{2} \varphi_\Lambda = F, \quad (32)$$

$$\dot{\varphi}_\Lambda + \frac{\zeta}{2} \varphi_\Lambda + \frac{\Omega_t^2 - \Omega_f^2}{2j\Omega_f} \varphi_\Lambda - \frac{3}{8\Omega_f^3} j |\varphi_\Lambda|^2 \varphi_\Lambda + \frac{\Theta}{2} \varphi_H = 0. \quad (33)$$

With structural damping ξ , equation (32) becomes

$$\dot{\varphi}_H + \left(\xi + \frac{1 - \Omega_f^2}{2j\Omega_f} \right) \varphi_H - \frac{\Theta}{2} \varphi_\Lambda = F. \quad (34)$$

5 Steady-State Cubic and Hysteresis

5.1 No Structural Damping

At steady state, eliminate φ_H from (32)–(33). The result can be written

$$A\varphi_\Lambda + C|\varphi_\Lambda|^2\varphi_\Lambda = B, \quad (35)$$

with

$$A = \frac{\zeta}{2} \frac{1 - \Omega_f^2}{2j\Omega_f} - \frac{(\Omega_t^2 - \Omega_f^2)(1 - \Omega_f^2)}{4\Omega_f^2} + \frac{\Theta^2}{4}, \quad (36)$$

$$B = -\frac{\Theta}{2} F, \quad (37)$$

$$C = -\frac{3(1 - \Omega_f^2)}{16\Omega_f^4}. \quad (38)$$

Taking the squared magnitude gives the real cubic relation

$$C^2 \rho^3 + 2 \operatorname{Re}(A) C \rho^2 + |A|^2 \rho = |B|^2, \quad \rho = |\varphi_\Lambda|^2. \quad (39)$$

This is the algebraic equation used to plot the hysteresis curve and determine stable/unstable branches.

5.2 With Structural Damping

With structural damping, define

$$D = \xi + \frac{1 - \Omega_f^2}{2j\Omega_f}. \quad (40)$$

The same cubic form (35) holds, but with

$$A = \frac{\zeta}{2}D + \frac{\Omega_t^2 - \Omega_f^2}{2j\Omega_f}D + \frac{\Theta^2}{4}, \quad (41)$$

$$B = -\frac{\Theta}{2}F, \quad (42)$$

$$C = -\frac{3}{8\Omega_f^3}jD. \quad (43)$$

6 Linear Stability of a Slow-Flow Branch

Let a steady solution be $(\varphi_{H,0}, \varphi_{\Lambda,0})$. Perturb it as

$$\varphi_H = \varphi_{H,0} + \delta_H, \quad \varphi_\Lambda = \varphi_{\Lambda,0} + \delta_\Lambda. \quad (44)$$

The perturbation dynamics can be written in the standard complex-conjugate linear form

$$\frac{d}{dT} \begin{bmatrix} \delta_H \\ \delta_\Lambda \end{bmatrix} = \mathbf{A} \begin{bmatrix} \delta_H \\ \delta_\Lambda \end{bmatrix} + \mathbf{B} \begin{bmatrix} \overline{\delta_H} \\ \overline{\delta_\Lambda} \end{bmatrix}. \quad (45)$$

For the undamped structural case,

$$\mathbf{A} = - \begin{bmatrix} \frac{1 - \Omega_f^2}{2j\Omega_f} & -\frac{\Theta}{2} \\ \frac{\Theta}{2} & \frac{\zeta}{2} + \frac{\Omega_t^2 - \Omega_f^2}{2j\Omega_f} - \frac{3j}{4\Omega_f^3}|\varphi_{\Lambda,0}|^2 \end{bmatrix}, \quad (46)$$

$$\mathbf{B} = - \begin{bmatrix} 0 & 0 \\ 0 & -\frac{3j}{8\Omega_f^3}\varphi_{\Lambda,0}^2 \end{bmatrix}. \quad (47)$$

The branch is stable if all eigenvalues of

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{bmatrix} \quad (48)$$

have non-positive real part. This is the stability check implemented in the ROM scripts when marking unstable portions of the hysteresis curve.

7 Power and Energy Accounting

The harvester output power is

$$P_1(t) = \frac{v_1^2(t)}{R_1} = \frac{1}{R_1}\dot{\lambda}_1^2. \quad (49)$$

In normalized variables, it is proportional to

$$P_{1,\text{norm}} = \zeta (\Lambda'_1)^2. \quad (50)$$

The prescribed modulator path does work on the beam through

$$P_2(t) = v_2(t)i_2(t), \quad (51)$$

where

$$i_2(t) = C_p \dot{v}_2(t) + \chi_{12} \dot{\eta}_1(t). \quad (52)$$

The capacitive term is reversible charging energy. The mechanically effective term is therefore

$$P_{2,\text{eff}}(t) = \chi_{12} v_2(t) \dot{\eta}_1(t). \quad (53)$$

The January 31 slides and `simulations/rom/corrected_fem_mode_rom.m` compare cumulative normalized harvested energy against modulation input. This is the right diagnostic for judging whether the nonlinear branch transition is useful, rather than merely producing a larger response amplitude.

8 Relation to Simulation Results

- **Initial single-mode ROM.** Hysteresis and branch selection under harmonic base excitation and modulation were checked in the older analytic mode scripts in `simulations/legacy_reference/`.
- **Continuous beam FEM.** The full beam model was used to test whether the ROM-predicted branch transition appears beyond the single-mode approximation. See `simulations/fem/continuous_harvester.m`.
- **Corrected mode assumption.** The FEM-informed mode shape and frequency replace the plain cantilever mode. See `simulations/fem/eigenanalysis_corrected_single_mode.m`.
- **Corrected ROM.** The main two-DOF model loads `FEM_modes_norm.mat` and should be compared against FEM. See `simulations/rom/corrected_fem_mode_rom.m`.
- **Power flow.** Harvested energy versus modulation input is discussed in the January 31 slides and implemented in the corrected ROM scripts under `simulations/rom/`.
- **Second-harmonic robustness.** Nonlinear and linear comparisons with a secondary base-excitation harmonic are archived under `simulations/robustness/second_harmonics/`.

9 Open Points for Collaborators

- The sign convention for χ_{11} , χ_{12} , and the FEM mode orientation should be kept consistent. The corrected MATLAB scripts use the negative of the FEM mode quantities loaded from `FEM_modes_norm.mat`.
- The current two-DOF ROM treats the modulator voltage as prescribed. If the modulator electronics are later modeled as a real circuit, it will add additional electrical states.
- The multi-mode Word note and `Old_sources/Multi_DOF/` are useful future directions, but the present mainline result is still the corrected single-mode ROM.
- Brownian or rough stochastic base motion needs care because the FEM boundary formulation uses prescribed acceleration. This was already flagged in the February 14 meeting.